Devashish Tupkary

Institute of Quantum Computing, University of Waterloo, Canada
tupkary.devashish@gmail.com
website, Google Scholar

RESEARCH INTERESTS

Quantum Key Distribution, Quantum Cryptography, Quantum Information, Classical Cryptography, Open Quantum Systems.
3 manuscripts in preparation, 12 peer-reviewed publications and 3 preprints under review.

PROFESSIONAL BACKGROUND

Postdoctoral Researcher — University of Montreal Sept 2026–Present

Postdoctoral Researcher — Institute for Quantum Computing May 2026–Aug 2026

Research Analyst — evolutionQ Feb 2024–Present

- Developing and evaluating cryptographic key management and routing algorithms for large-scale QKD networks.
- Leading consulting engagements and security reviews for client QKD implementations, including assessment of classified design documents. Responsible for project planning, budgeting, negotiation, and delivery and funding applications.

ACADEMIC BACKGROUND

Ph.D. Physics (Quantum Information), GPA : 92/100 Jan 2023-April 2026
Institute for Quantum Computing, University of Waterloo
Mike and Ophelia Lazaridis Fellowship.

- Research on rigorous security analysis of practical quantum key distribution protocols. Thesis Title : Rigorous Security Proofs for Practical Quantum Key Distribution.
- Relevant Coursework (Quantum Information and Cryptography): Quantum Error-correction and Fault-Tolerence, Quantum Information Processing, Implementation of Quantum Information Processing, Theory of Quantum Information, Applied Quantum Cryptography, Applied Cryptography, Mathematics of Public Key Cryptography, Privacy Data and Security.
- Contributor to Open QKD Security software package for numerical QKD analysis.

M.Sc. Physics (Quantum Information), GPA : 96/100 Sept 2020- Sept 2022
Institute for Quantum Computing, University of Waterloo
Thesis Title : Improved Keyrates for Quantum Key Distribution from two-way Classical Communication

B.Sc. (Research) Physics, CGPA : 9.2/10 2016-2020
Indian Institute of Science

- Graduated First Class with Distinction.

- Relevant Coursework (Quantum Information and Cryptography): Quantum Mechanics 1, Quantum Mechanics 2, Quantum Computation, Quantum Measurements and Control, Quantum Information Theory, Information Theory, Theoretical Foundations of Cryptography.

FELLOWSHIPS

- Mike and Ophelia Lazaridis Fellowship - Fellowship during PhD, includes tuition and a yearly stipend of 36000 CAD. April 2023-April 2026
- Long Term Visiting Student Program, International Centre for Theoretical Studies - Tata Institute of Fundamental Research - Among a dozen students selected for a year-long funded research project. 2019
- Kishore Vaigyanik Protsahan Yojana (KVPY-SA Fellowship). *Rank 39* - Fellowship during undergraduate degree, includes a yearly stipend of 80000 INR. 2016-2020

PAPER AWARDS

- Cryptoworks21 Distinguished Student Paper Award - For [P15]. 2025
- Works [P15,P13] (combined) and [P14,P12] (combined) shortlisted for best student theory submission at QCrypt 2024. 2024

INVITED TALKS

- Invited Tutorial Talk at QCrypt 2025 on "Security Proofs for Decoy-State BB84 and Their Performance" based on [P10]. 26 August 2025
- Lecture at Cryptoworks21 program, University of Waterloo, on "An Introduction to Quantum Key Distribution." 22 July 2025
- Invited Talk at CQIQC Seminar, University of Toronto, on "Lindbladians Obeying Local Conservation Laws and Showing Thermalization" based on [P17,P18]. 10 February 2023

CONTRIBUTED TALKS

- Contributed Talk at QCrypt 2026 on "A rigorous and complete security proof of decoy-state BB84 quantum key distribution". Based on [P5]. August 6
- Contributed Talk at TQC 2025 on "Phase error rate estimation in QKD with imperfect detectors". Based on [P13]. 16 Sept 2025
- Contributed Talk at QCrypt 2024 on "Security Proof for Variable-Length Quantum Key Distribution" and "Variable-Length QKD Security Proof for Imperfect Detectors through Phase-Error Estimation". Based on [P15,P13]. 2 September 2024

SEMINARS

- Talks at University of Ottawa, University of Montreal, École de technologie supérieure, University of Sherbrooke on "Rigorous Security Proofs for practical Quantum Key Distribution. Based on PhD Thesis. March 2026
- Talks at QEYSSAT User Investigation Team (QUINT) meeting. 17th Feb 2026
- Talk at IISc Bengaluru, IISER Pune on "Extending Quantum Key Distribution security proofs to practical detectors". Based on [P13,P7,P9]. 11th,12th Sept 2025

- Contributed Talk at QKD Security Proof workshop on “Classical Authentication in QKD”. Based on [P4]. 23rd Sept 2025
- Talk at IIT Bombay on “An Introduction to Quantum Key Distribution”. 1st Sept 2025
- Talk at ICTS-TIFR on “ A brief introduction to Quantum Key Distribution”. 4th Jan 2024
- Contributed Talk at QKD Security Proof workshop on “Phase error estimation with basis-efficiency mismatch”. Based on [P13]. 31st Sept 2024
- Contributed Talk at QKD Security Proof workshop on “Postselection technique” and “Variable-length QKD.” Based on [P15,P14] 12th,13th Sept 2023
- Contributed Talk at QKD Security Proof workshop on “Improved processing of classical data in QKD”. 21st Sept 2022

STUDENT AWARDS and COMPETITIONS

- Madhava Mathematical Competition. *Rank 6* - A nationwide mathematics competition for undergraduate students. 2017
- National Bal Shree Honour for Creative Scientific Innovation - One of the three highest national honours for children by the Government of India. 2016
- IIT-JEE (Advanced). *Rank 312* - A nationwide entrance exam for college admissions to the Indian Institute of Technology (IITs). 2016
- Indian National Mathematics Olympiad - Ranked among the top 70 nationwide. 2016
- Qualified Indian National Physics Olympiad - Ranked among the top 35 nationwide. 2016
- National Talent Search Examination - Among 1000 students awarded the NTSE Scholarship nationally in each year. 2012-2016

SUPERVISION

- Supervised Shifan He, undergraduate research project. 2026
- Supervised Jerome Wiesemann, research project leading to [P2]. 2025–2026
- Supervised Zhiyao Wang, research project at IQC Waterloo, leading to [P7]. 2024–2025
- Supervised Jerome Wiesemann, research project leading to [P11]. 2024–2025
- Supervised Soumyadeep Sarma, Bachelor’s thesis at ICTS–TIFR, Bengaluru, leading to [P6]. 2024–Present

TEACHING

- TA for QIC890, Applied Quantum Cryptography Winter 2026
- TA for PHYS115, Mechanics Fall 2021
- TA for PHYS242, Electricity and Magnetism Spring 2021, 2023

POSITIONS OF RESPONSIBILITY

- Co-Organizer of the QKD Security Proof Workshop at IQC Waterloo. 2022,2023,2024
- Co-Organizer of the weekly IQC Student Seminar Series. 2023-2024

PUBLICATIONS

[P1] ***Beyond MEAT with detector correlations, source correlations and iterative sifting***.
Devashish Tupkary, Amir Arqand, *Manuscript in preparation*.

[P2] ***Security framework for practical quantum key distribution with imperfect devices***.
Jerome Wiesemann, John Burniston, Norbert Lütkenhaus, Devashish Tupkary, *Manuscript in preparation*.

[P3] ***Phase Error Estimation for Dynamically Modulated Single Photon BB84 with Device Imperfections***.
Aodhán Corrigan, Koray Kaymazlar, Zhiyao Wang, Devashish Tupkary, Lucas Rickert, Daniel Vajner, Martin von Helversen, and Tobias Heindel, *Manuscript in preparation*.

[P4] ***Authentication in Security Proofs for Quantum Key Distribution***.
Devashish Tupkary, Shlok Nahar, Ernest Y.Z-Tan, arXiv link. Under review at IEEE Transactions in Information Theory.

[P5] ***A rigorous and complete security proof of decoy-state BB84 quantum key distribution***.
Devashish Tupkary, Shlok Nahar, Amir Arqand, Ernest Y.Z-Tan, Norbert Lütkenhaus, arXiv link. Under review at Quantum.

[P6] ***Semidefinite Programming for understanding limitations of Lindblad Equations***.
Soumyadeep Sarma, Manas Kulkarni, Archak Purkayastha, Devashish Tupkary, Phys. Rev. A .

[P7] ***Phase error estimation for passive detection setups with imperfections and memory effects***.
Zhiyao Wang, Devashish Tupkary, Shlok Nahar, arXiv link. Under review at NPJ Quantum Information.

[P8] ***Imperfect detectors for adversarial tasks with applications to quantum key distribution***.
Shlok Nahar, Devashish Tupkary, Norbert Lütkenhaus, Quantum 10, 2044 (2026).

[P9] ***Security of quantum key distribution with source and detector imperfections through phase-error estimation***.
Guillermo Currás-Lorenzo, Margarida Pereira, Shlok Nahar, Devashish Tupkary, NPJ Quantum Information (2026).

[P10] ***QKD security proofs for decoy-state BB84: protocol variations, proof techniques, gaps and limitations***.
Devashish Tupkary, Ernest Y-Z Tan, Shlok Nahar, Lars Kamin, Norbert Lütkenhaus, Reviews of Modern Physics.

[P11] ***A consolidated and accessible security proof for finite-size decoy-state quantum key distribution***.
Jerome Wiesemann, Jan Krause, Devashish Tupkary, Davide Rusca, Nino Walenta, Norbert Lütkenhaus, Quantum 10, 2037 (2026).

- [P12] ***Improved finite-size effects in QKD protocols with applications to decoy-state QKD***.
  Lars Kamin, Devashish Tupkary, Norbert Lütkenhaus, Phys. Rev. Research 8, 013332.
- [P13] ***Phase error rate estimation in QKD with imperfect detectors***.
  Devashish Tupkary, Shlok Nahar, Pulkit Sinha, Norbert Lütkenhaus, Quantum 9, 1937 (2025).
- [P14] ***Postselection technique for optical prepare-and-measure QKD protocols***.
  Shlok Nahar, Devashish Tupkary, Yuming Zhao, Norbert Lütkenhaus, Ernest Y.-Z. Tan, PRX Quantum 5, 040315.
- [P15] ***Security Proof for Variable-Length Quantum Key Distribution***.
  Devashish Tupkary, Ernest Y.-Z. Tan, Norbert Lütkenhaus, Phys. Rev. Research 6, 023002.
- [P16] ***Using Cascade in Quantum Key Distribution***.
  Devashish Tupkary, Norbert Lütkenhaus, Phys. Rev. Applied 20, 064040.
- [P17] ***Searching for Lindbladians obeying local conservation laws and showing thermalization***.
  Devashish Tupkary, Abhishek Dhar, Manas Kulkarni, and Archak Purkayastha, Phys. Rev. A 107, 062216.
- [P18] ***Fundamental Limitations in Lindblad Descriptions of systems weakly coupled to baths***.
  Devashish Tupkary, Archak Purkayastha, Abhishek Dhar, Manas Kulkarni. Phys. Rev. A 105, 032208.

RECOMMENDATION LETTERS

The following individuals have agreed to provide letters of recommendation on my behalf:

- **Prof. Norbert Lütkenhaus**
  Professor and Executive Director, Institute for Quantum Computing and Department of Physics
  University of Waterloo, Canada
  nlutkenhaus.office@uwaterloo.ca
- **Prof. Michele Mosca, FRSC**
  Professor, Institute for Quantum Computing and Department of Combinatorics & Optimization
  University of Waterloo, Canada
  CEO and Co-Founder, evolutionQ Inc.
  michele.mosca@uwaterloo.ca
- **Prof. Dagmar Bruss**
  Professor of Theoretical Physics
  Heinrich Heine University Düsseldorf, Germany
  dagmar.bruss@uni-duesseldorf.de
- **Prof. Manas Kulkarni**
  Faculty Member, International Centre for Theoretical Sciences (ICTS-TIFR)
  Bangalore, India
  manas.kulkarni@icts.res.in